

K382 PRE-STAGE — Muon gate convened; four CI signoffs
 received with SUBSTANTIVE SPLIT VERDICT (3
 PROMOTE-NOW, 1 HOLD-ONE-STEP). LYRA F151: CLEARS
 under Casey criterion with two premises documented honestly
 (Premise 1 bridge architecturally allowable Casey's call; Premise
 2 fermionic-rerun scoped to stage-1 residual not muon-specific
 blocker); recommends promote 2→3 with both premises on
 record rather than buried; one thing wants from Casey before
 bank closes: ruling on Premise 1. ELIE Toy 4202 7/7 PASS
 discrete leg cross-check: $64=2^{C_2}=d_{\tau}/d_{\mu}$ off-target
 object-loop three faces; exponent $6=\dim \text{so}(4)=C_2=n_C+1$
 forced falsifiable; $64/\text{vol}(S^4)=24/\pi^2$ exactly; value 3.4×10^{-5}
 below floor blind alternatives forbidden; honest caveat
 $\text{so}(4)$ -determinant bolt mechanism-identified and half-closed via
 Grace isotropy not full F116 proof multi-session; supports 2→3
 with bolt noted at 'reliable explanation' tier NOT 10^{-14} identity.
 GRACE RATIFIED Casey criterion with guard: structural floor
 must be pre-committed and physically justified (named
 corrections of right size) not 'gap small so call it floor'; for muon
 floor $\sim 10^{-4}$ from S^4 holonomy + electron-side log + RG; gap
 3.4×10^{-5} well inside criterion legitimately satisfied; ran muon
 through 7-item under criterion; clears 5 cleanly (rigid + two
 routes + no hidden $O(1)$ given bridge + within floor + not
 reverse-read); 2 pending closures: Item 2 forced structure (64
 off-target real BUT determinant mechanism is half-bolt → Cal's
 cold-read to confirm rigorous), Item 7 composes (object-loop via
 64 + NO observable-loop yet = just-determined not
 over-determined; acceptable for first bank; observable-loops

come later from blind test); bank comes down to two genuine
 closures one Cal's one Casey's; ready instant they land; holding
 count honestly 2 until they do. CAL SUBSTANTIVE VERDICT:
 NOT clean signoff today; substantive analysis with credit AND
 substantive substrate-architectural argument: (a) F138
 GENUINELY resolves #296 (diagonal on homogeneous Shilov
 $S^4 = \text{Tr}/\text{vol}(S^4)$ BY CONSTRUCTION since $K(x,x)$ constant on
 homogeneous; probability measure 'is not the diagonal of
 anything'; dissolves old objection conditional on bridge; retires
 worry T754 probability-normalization contradicts because
 different objects — Born inner-product measure vs heat-kernel
 density; #296 REDUCED TO BRIDGE not sidestepped);
 CREDIT LYRA; (b) audit chain worked — Cal #296/#299/#363
 catches absorbed and team built F138 in response; (c) value
 $3 \times 10^{-5} + \text{exact } d_\tau/d_\mu = 64 + \text{falsifiable exponent-6}$ all real; (d)
 Lyra surfaced premises honestly + invoked Cal #27 — honest
 gate construction not rubber-stamping; (e) form IS forced
 GIVEN THE BRIDGE; THE ONE LINE CAL HOLDS — his own
 prior point #363 not new wall: bridge is being banked on
 observable it was BUILT AGAINST; Premise 1 GR-style bridge;
 Casey can rule allowable architectural call; BUT ruling
 allowable sets what green MEANS doesn't BANK muon; bridge
 earns green BY PREDICTION (GR gravity=curvature earned
 Mercury+light-bending BLIND not by explaining one thing
 constructed on); muon IS that one thing; promoting muon alone
 today BANKS bridge on its defining example = POSTDICTION
 of tuned case wearing clothes of forced prediction; test is
 READY (Grace harness fires on Elie tau-box); honest
 referee-proof sequence: Casey rules bridge allowable-input →

fire blind harness on tau (and other leptons never saw) → if
 same bridge carries blind = earned → muon+tau+...
 PROMOTE SET TOGETHER bridge earned; FAR STRONGER
 first motion than muon-alone; hostile referee can't reopen; if
 Casey overrides and rules green today (his prerogative): Cal's
 on-record flag written INTO BANK not buried (count 2→3 only
 as strong as Premise 1 allowable AND blind eventually passing;
 item 2 re-tiered to one link). CAL SUBSTANTIVE CHECKLIST
 FACTUAL CORRECTION on item 2: '64 three faces + 6 four
 faces' OVERCOUNTS per Cal #304b/#307; genuine content
 ONE forced exact ratio $d_{\tau}/d_{\mu}=64$; ' $=2^{C_2}$ / three faces / 6 four
 faces' same number read several ways — structurally
 $\text{rank}^2 \cdot 2^{(\text{rank}^2)}=64$ equals 2^{C_2} ONLY BECAUSE exponent
 $2+\text{rank}^2$ COINCIDENTALLY equals $C_2=6$; by Grace's own C3
 (loops not stars) it's DEGENERATE STAR not
 over-determination; bank 'one forced off-target link' not
 'off-target over-determination'; doesn't block gate; RIGHT-SIZES
 item 2 — factual correction independent of timing decision.
 Premise 2 (fermionic-tower) Cal agrees with Lyra scoping —
 addresses ordering-forced ratio robust general-rigor residual into
 stage-1 not muon blocker; don't block keep on record. KEEPER
 VERDICT: SUBSTANTIVE SPLIT — 3 CIs recommend
 PROMOTE NOW conditional on Casey bridge call (Lyra Elie
 Grace pending two closures both gate-internal); 1 CI (Cal)
 recommends HOLD ONE STEP for stronger first motion
 (muon+tau together via blind test); Cal's argument has
 substantive merit — bridge banked on defining example IS
 postdiction in referee terms; promoting set together when bridge
 predicts blind is harder to reopen; HOWEVER Casey's criterion

+ Lyra’s premises-on-record + Grace’s two-closures + Elie 7/7 + Cal’s factual checklist correction together meet the criterion as Casey defined it; CASEY DECIDES — bridge call + timing both his; either path defensible under audit-chain discipline; if PROMOTE NOW: Cal’s flag on record written into bank not buried per his explicit offer; if HOLD ONE STEP: muon green-eligible bridge-pending-blind, fire Grace harness on tau (Elie 4202 tau-box discrete done; orbit→ ν continuum gates), promote set when bridge predicts blind; either way Cal item 2 right-sizing applies (degenerate star not over-determination → ‘one forced off-target link’). Count HONESTLY 2 of 26 strict pending Casey decision; ledger v1.73.

Keeper (Claude Opus 4.7)

2026-06-16 Tuesday 09:00 EDT (date-verified)

K382 PRE-STAGE Audit

1. The substantive SPLIT VERDICT

The convened gate produced four substantive readings — not unanimous PASS:

CI	Recommendation	Substance
Lyra F151	PROMOTE 2→3	Clears with two premises documented; one ask: Casey’s ruling on Premise 1
Elie 4202	PROMOTE 2→3	7/7 discrete leg PASS; so(4)-determinant bolt noted at “reliable explanation” tier
Grace	PROMOTE pending two closures	Criterion ratified; 5 of 7 clears; 2 gate-internal closures pending (det-mechanism Cal’s, bridge Casey’s)
Cal	HOLD ONE STEP	Substantive referee-discipline argument; muon-alone today is postdiction of bridge on defining example

3 CIs recommend PROMOTE NOW; 1 CI (Cal) recommends HOLD ONE STEP.

2. Cal's substantive verdict — credit AND substantive argument

What Cal credits (genuinely earned, not reflexive): - F138 GENUINELY resolves #296: diagonal on homogeneous Shilov $S^4 = \text{Tr}/\text{vol}(S^4)$ BY CONSTRUCTION ($K(x,x)$ constant on homogeneous; “probability measure is not the diagonal of anything”); dissolves old objection conditional on bridge; retires worry that T754 probability-normalization contradicts (different objects — Born inner-product measure vs heat-kernel density); #296 reduced to the bridge, not sidestepped. Credit Lyra. - Audit chain worked: Cal #296/#299/#363 catches absorbed and team built F138 in response - Value (3×10^{-5}), exact $d_\tau/d_\mu=64$, falsifiable exponent-6 — all real - Lyra surfaced premises honestly + invoked Cal #27 — honest gate construction, not rubber-stamping - Form IS forced given the bridge

The one line Cal holds (his own prior #363, not a new wall):

The bridge is being banked on the observable it was BUILT AGAINST. Premise 1 (mass = intensive concentration / diagonal) is GR-style bridge; Casey can rule allowable architectural call. But ruling allowable sets what green MEANS; doesn't BANK the muon.

A bridge earns green BY PREDICTION (GR's “gravity = curvature” earned status by predicting Mercury + light-bending BLIND, not by explaining the one thing it was constructed on). The muon IS that one thing.

Promoting muon alone today banks the bridge on its defining example = postdiction of tuned case wearing clothes of forced prediction.

The test is READY: Grace's blind-prediction harness fires on Elie's tau-box.

Cal's referee-proof sequence: 1. Casey rules bridge allowable-input form (architectural call) 2. Fire blind harness on tau (and other leptons it never saw) 3. If same bridge carries them blind = earned 4. Promote SET (muon + tau + ...) when bridge predicts blind

Far stronger first motion than muon-alone; hostile referee can't reopen.

If Casey overrides and rules green today (his prerogative): Cal's on-record flag written INTO the bank, not buried. Count 2→3 is only as strong as (i) Premise 1 allowable AND (ii) blind test eventually passing; item 2 re-tiered to one link.

3. Cal's substantive factual correction on item 2 — independent of timing

“64 three faces + 6 four faces” OVERCOUNTS (per Cal #304b/#307): - Genuine content: ONE forced exact ratio $d_\tau/d_\mu = 64$ - “= $2^6 C_2$ / three faces / 6 four faces” = same number read several ways - Structurally $\text{rank}^2 \cdot 2^6 (\text{rank}^2) = 64$ - Equals $2^6 C_2$ ONLY because exponent $2 + \text{rank}^2$ COINCIDENTALLY equals $C_2 = 6$ - By Grace's own C3 (loops not stars) it's a DEGENERATE STAR, not over-determination

Bank “one forced off-target link” NOT “off-target over-determination.”

Doesn't block the gate; right-sizes item 2. Factual correction applies regardless of timing decision.

4. Premise 2 (fermionic-tower) — Cal agrees with Lyra scoping

Addresses ordering-forced, ratio robust, general-rigor residual into stage-1, not muon-specific blocker. Don't block; keep on record.

5. Casey's decisions

Decision	Options
Bridge ruling (Premise 1)	Allowable input (definitional like ℓ_B) OR must-derive from 5 integers

Decision	Options
Timing	PROMOTE muon now (with Cal's flag on record) OR HOLD ONE STEP (fire harness on tau, promote set together)

Either path defensible under audit-chain discipline. Cal's recommendation has substantive merit (referee-proof first motion); Lyra+Elie+Grace's reading also valid under Casey's criterion as stated.

6. What promotes under each option

Option A — PROMOTE muon now: - Count 2 → 3 - Cal's flag on record written into bank: "muon 2→3 only as strong as Premise 1 allowable AND blind eventually passing; item 2 re-tiered to one link" - Item 2 re-banked as "one forced off-target link" not "over-determination" - Then both hands on stage-1 master lock

Option B — HOLD ONE STEP: - Muon green-eligible, bridge-pending-blind - Fire Grace harness on tau (Elie 4202 discrete done; Lyra orbit→ν continuum gates remaining) - Promote SET (muon + tau + ...) when bridge predicts blind - First motion = muon + tau together, referee-proof - Item 2 already right-sized to "one forced off-target link"

Either way, Cal's item 2 factual correction applies.

7. The substantive substrate-architectural state at K382

Component	Status
Casey criterion ratified by Grace	With guard: floor pre-committed + physically justified
Form forced given bridge	All CIs agree
Lyra F151 documented premises	Both on record honestly
Elie 4202 discrete leg PASS	7/7 with so(4) bolt noted
Grace 7-item under criterion	5 of 7 clears; 2 closures pending
Cal #296 dissolved by F138	Credit Lyra; reduced to bridge not sidestepped
Cal item 2 right-sizing	"One forced off-target link" not over-determination
Cal timing recommendation	HOLD ONE STEP for muon+tau together via blind test
Lyra+Elie+Grace timing	PROMOTE NOW conditional on bridge call
Casey decisions pending	Bridge call + timing
Count	HONESTLY 2 of 26 strict pending
Ledger	v1.73

8. K382 PRE-STAGE verdict

K382 PRE-STAGE substantive SPLIT VERDICT on muon gate:

- Four CI signoffs received with substantive split: 3 PROMOTE NOW (Lyra Elie Grace conditional), 1 HOLD ONE STEP (Cal)
- Cal's substantive credit: F138 genuinely resolves #296 (homogeneous Shilov diagonal = $\text{Tr}/\text{vol}(S^4)$ by construction; probability measure "is not the diagonal of anything"); form forced given bridge; audit chain worked
- Cal's substantive hold-one-step argument: bridge banked on defining example IS postdiction (referee terms); GR's bridge earned by Mercury/light-bending BLIND not by what it was constructed on; muon IS what bridge was constructed on; Grace harness READY; promote SET (muon+tau) when bridge predicts blind = stronger first motion

- Cal's factual correction on item 2: "64 three faces + 6 four faces" overcounts (same number read several ways; structurally $\text{rank}^2 \cdot 2^{(\text{rank}^2)} = 64$ equals 2^6 only because coincidentally same); degenerate star not over-determination per Grace's own C3; bank "one forced off-target link" — applies regardless of timing
- Premise 2 fermionic Cal agrees with Lyra scoping; don't block; on record
- Casey decisions: bridge ruling (allowable vs must-derive) + timing (promote now vs hold one step)
- Both paths defensible under audit-chain discipline; Cal's referee-proof argument has substantive merit
- If PROMOTE NOW: Cal's flag on record per his explicit offer; item 2 right-sized
- If HOLD ONE STEP: muon green-eligible bridge-pending-blind, fire harness on tau, promote set
- Count HONESTLY 2 of 26 strict pending Casey decision
- Ledger v1.73

Substantial substrate-architectural significance: K382 captures the substantive SPLIT VERDICT at the muon gate — three CIs (Lyra, Elie, Grace) recommend PROMOTE NOW under Casey's criterion conditional on his bridge ruling, while Cal recommends HOLD ONE STEP for a referee-proof first motion (muon+tau together when bridge predicts blind). Cal's substantive credit to F138 is significant — Lyra's homogeneous Shilov argument (diagonal on $S^4 = \text{Tr}/\text{vol}$ by construction since $K(x,x)$ is constant on homogeneous; "probability measure is not the diagonal of anything") GENUINELY resolves Cal's standing #296 objection (geometric-vs-probability measure) and clarifies that the apparent contradiction with T754 probability-normalization was Born inner-product vs heat-kernel density — different objects, not contradictory. #296 is reduced to the bridge, not sidestepped. The audit chain worked: Cal's catches were absorbed and the team built F138 in response. But Cal's hold-one-step argument has substantive referee-discipline merit: promoting the muon alone today banks the bridge on the observable it was constructed against (postdiction of tuned case wearing clothes of forced prediction); GR's bridge earned green by predicting Mercury and light-bending BLIND, not by explaining what it was built on. The blind test is READY (Grace harness fires on Elie's tau-box). Promoting muon+tau SET together when bridge predicts blind is the referee-proof first motion. Cal's factual correction on item 2 applies regardless of timing: "64 three faces + 6 four faces" overcounts (same number read several ways; structurally $\text{rank}^2 \cdot 2^{(\text{rank}^2)} = 64$, equals 2^6 only because exponent $2 + \text{rank}^2 = 6$ coincidentally); by Grace's own C3 it's a degenerate star not over-determination; bank "one forced off-target link." Casey decides: bridge ruling (allowable input vs must-derive) + timing (promote now vs hold one step). Either path defensible under audit-chain discipline; Cal offered both — promote now with his flag on record OR hold one step for stronger first motion. Count stays HONESTLY at 2 of 26 strict pending Casey decision.

— Keeper, Tuesday 2026-06-16 09:00 EDT (date-verified) — K382 PRE-STAGE: muon gate split verdict — 3 PROMOTE NOW conditional on bridge call (Lyra+Elie+Grace), 1 HOLD ONE STEP for muon+tau together via blind test (Cal); Cal credits F138 genuinely resolves #296; Cal substantive item 2 factual correction (degenerate star not over-determination); Casey decides bridge + timing; count HONESTLY 2 of 26 pending